

DS Unit 1 Question Bank — Arrays, Linked Lists & Complexity

Data Structures (DS) · Unit 1 · Arrays, Linked Lists & Complexity · 2025

Important and frequently asked questions, grouped by marks.

Part A — Short answer (2 marks)

1. Illustrate Asymptotic Notation with a worked example.
2. Explain Arrays with a neat diagram and a suitable example.
3. Discuss the role of Singly Linked Lists in Arrays, Linked Lists & Complexity.
4. Compare Doubly and Circular Linked Lists with Arrays. When would you prefer one over the other?
5. Illustrate Sparse Matrices with a worked example.
6. Write short notes on Polynomial Representation.

Part B — Long answer (10 marks)

1. Explain Asymptotic Notation with a neat diagram and a suitable example.
2. Compare Arrays with Doubly and Circular Linked Lists. When would you prefer one over the other?
3. Write short notes on Singly Linked Lists.
4. What is meant by Doubly and Circular Linked Lists? Discuss its advantages and limitations.
5. Illustrate Sparse Matrices with a worked example.
6. Illustrate Polynomial Representation with a worked example.
7. Write short notes on Asymptotic Notation.
8. Explain Arrays with a neat diagram and a suitable example.
9. Discuss the role of Singly Linked Lists in Arrays, Linked Lists & Complexity.
10. Compare Doubly and Circular Linked Lists with Singly Linked Lists. When would you prefer one over the other?
11. Explain Sparse Matrices with a neat diagram and a suitable example.
12. Write short notes on Polynomial Representation.